Evidence Before This Study — Literature Search and Synthesis (NTD Burden and Funding Cuts)

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Search strategy

We searched PubMed from database inception to 21 October 2026 with six keyword strings, one per evidence theme. We supplemented this with grey literature: the WHO global report on NTDs, the World Malaria Report, and IHME Financing Global Health. No language restrictions were applied at the search stage.

Theme	Search string (PubMed)	Limit	Records
A. Global, regional and national burden	("neglected tropical disease(s)") AND ("Global Burden of Disease" OR "disability-adjusted life years" OR DALY*)	Title/abstract	206
B1. Inequality	("neglected tropical diseases" OR malaria [title]) AND (inequalit* OR "concentration index" OR "slope index" OR "socio-demographic index") AND ("Global Burden" OR GBD)	Title/abstract	29
B2. Spatial clustering	("neglected tropical diseases") AND (spatial OR geospatial OR Moran OR hotspot* OR cluster*) AND (global OR countries OR Africa)	Title/abstract	176
C. Financing and outcomes	("neglected tropical diseases" OR malaria) AND ("development assistance for health" OR "external financing" OR "donor funding" OR "foreign aid" OR "health aid" OR "official development assistance")	Title/abstract	163

Theme	Search string (PubMed)	Limit	Records
D. 2024–2026 funding-cut projections	(USAID OR "funding cuts" OR defunding OR "aid cuts" OR "funding reduction" OR "President's Malaria Initiative") AND (malaria OR "neglected tropical") AND (projected OR projection OR modelling OR forecast)	Jan 2024 – Oct 2026	99
E. Aid, mortality and donor transition	("development assistance for health" OR "health aid" OR "donor transition") AND (mortality OR burden) AND (panel OR "fixed effects" OR "difference-in-differences" OR countries)	Title/abstract	94

Eligibility. We included studies that met at least one of the following criteria:

- multi-country or global analyses of NTD or malaria burden with temporal trends;
- cross-country analyses of inequality or spatial clustering of that burden;
- quantitative cross-country analyses linking donor or external financing to NTD, malaria or all-cause mortality outcomes;
- models projecting the health consequences of the 2024–2026 aid reductions;
- studies of the domestic replacement or displacement of health aid.

We excluded single-country studies (except a few landmark national analyses, flagged as such), clinical and laboratory studies, protocols, and commentaries without data. Of 767 records identified, 63 duplicates were removed, leaving 704. Six records with no retrievable metadata were excluded. Of 698 titles screened, 602 were excluded. Of 96 abstracts assessed, 20 were excluded: 10 commentaries or reviews without data, 3 superseded analyses, 6 off scope and 1 single-region study of a single disease. Of the 76 eligible studies, the 44 most relevant and recent are tabulated: 10 on burden, 10 on inequality and spatial patterns, 12 on financing and outcomes, 5 on projections of the cuts and 7 on domestic financing. Two further targeted searches found commentary on the 2025 cuts. The first covered Lancet journals in 2025–2026 (56 records); the second covered NTDs and funding cuts in 2025–2026 (8 records) and found no primary study projecting NTD health losses.

Evidence map

The 44 tabulated studies fall into five themes. Most global analyses describe NTD and malaria burden with GBD 2019 or 2021 data. None of the tabulated studies combines spatial clustering, socioeconomic inequality and financing dependence. Every study linking financing to NTD outcomes measures mortality rather than DALYs.

Table 1

Characteristics and Findings of Key Studies Identified in the Literature Search, by Theme

Study	Design and data	Scope	Key findings	Limitation relative to this study
Theme A: Global, regional and national burden and trends				
Murray et al. (2014), *Lancet*	GBD 2013; HIV, TB and malaria incidence and mortality, 1990–2013	188 countries	Malaria peaked at 232 million cases in 2003 and 1.2 million deaths in 2004; child malaria deaths in sub-Saharan Africa fell 31.5% after 2004	Earlier GBD round; no NTDs; no inequality, spatial or financing analysis
Martins-Melo et al. (2018), *PLoS NTD* [national]	GBD 2016 subnational, 12 NTDs, 1990–2016	Brazil, 27 states	Age-standardised NTD DALY rate fell 45.7%; Chagas disease leading cause	Single country; no financing
Lin et al. (2022), *J Travel Med*	GBD 2019; estimated annual percentage change (EAPC), 1990–2019	204 countries	Age-standardised NTD incidence rose while mortality and DALY rates fell	No malaria; descriptive only

Study	Design and data	Scope	Key findings	Limitation relative to this study
GBD 2021 Causes of Death Collaborators (2024), *Lancet*	GBD 2021; 288 causes, 1990–2021	204 countries	Malaria among causes increasingly concentrated in populations holding under half of the world population	All-cause scope; no NTD-specific inequality or financing
Zheng et al. (2025a), *J R Soc Med*	GBD 2019; malaria and NTDs, ages 0–19	204 countries	43.9 million DALYs in 2019; malaria > 93% of DALYs under age 5; burden inversely related to SDI	GBD 2019; children only; no formal inequality index
Zheng et al. (2025b), *J Adv Res*	GBD 2021; NTDs and malaria, 1990–2021	204 countries	DALYs fell 18.1%, from 87.4 million to 71.6 million; prevalence fell 58.1%	GBD 2021; descriptive SDI gradient only
GBD 2023 Chagas Disease Collaborators (2025), *Lancet Infect Dis*	GBD 2023; Chagas disease, 1990–2023	204 countries	10.5 million prevalent cases in 2023; age-standardised mortality fell 72.5% since 1990	Single NTD; no financing

Study	Design and data	Scope	Key findings	Limitation relative to this study
GBD 2023 Disease and Injury and Risk Factor Collaborators (2025), *Lancet*	GBD 2023; 375 causes, 1990–2023	204 countries	Age-standardised DALY rate fell 12.6% (2010–2023); communicable, maternal, neonatal and nutritional DALYs fell from 874 to 681 million; authors warn aid cuts threaten gains	All causes; no NTD-specific spatial, inequality or financing analysis
GBD 2023 Causes of Death Collaborators (2025), *Lancet*	GBD 2023; 292 causes of death	204 countries	Mean age at death in sub-Saharan Africa 38.0 (female) and 35.6 (male) years in 2023	Mortality only
Yin et al. (2026), *Trans R Soc Trop Med Hyg*	GBD 2023; infectious-disease deaths and EAPC	5 BRICS countries	Infectious deaths fell from 5.01 to 2.57 million (1990–2023)	Five countries; deaths only
Theme B: Inequality and spatial patterns				
Stolk et al. (2016), *PLoS NTD*	GBD 2010; between-country inequality by income group, 9 NTDs	Global	Low- and lower-middle-income countries held 69% of NTD burden in 1990 and 81% in 2010	Income groups rather than SDI; no index or spatial statistics; no malaria

Study	Design and data	Scope	Key findings	Limitation relative to this study
Lo et al. (2019), *Lancet Glob Health*	DHS spatiotemporal study of deworming coverage, 2003–2017	50 countries	30 of 31 countries reporting > 75% coverage showed coverage concentrated among wealthier groups	Coverage, not burden; no financing
Emadi et al. (2021), *BMC Public Health*	GBD 2019; concentration index by Human Development Index	204 countries	Concentration index for NTD and malaria DALYs −0.69, the most pro-poor communicable-disease group	Single year; no trend, spatial or financing analysis
Liu et al. (2023), *Glob Health Res Policy*	GBD 2019; joinpoint and SDI regression	China and 10 ASEAN countries	U-shaped association of incidence and mortality with SDI	Regional; GBD 2019
Apeagyei et al. (2024a), *Malar J*	Subnational Gini indices and malaria spending, 2010–2020	40 endemic countries	Incidence Gini 0.06–0.73; higher malaria spending associated with lower within-country inequality	Malaria only; within-country inequality
Lin et al. (2025), *Sci Rep*	GBD 2021; inequality and frontier analysis with projection to 2041	204 countries	Age-standardised DALY rate fell from 1,506.5 to 1,020.3 per 100,000 (1990–2021); western sub-Saharan Africa held 51.2% of burden	No financing; no spatial autocorrelation

Study	Design and data	Scope	Key findings	Limitation relative to this study
Khaki et al. (2025), *PLoS NTD*	Model-based geostatistics, soil-transmitted helminth surveys	35 SSA countries	Spatial correlation range 1.1–3,027 km; strong between-country heterogeneity	Single NTD prevalence; no financing
Namubiru (2025), *Malar J*	Spatial Durbin and fixed-effects models, Driscoll–Kraay SE, 2010–2022	38 SSA countries	Malaria cases and deaths clustered every year (Moran′s I); aid per capita positively associated with burden, consistent with targeting	Malaria only; SSA only; WHO/MAP outcomes, not GBD DALYs
Feng et al. (2025), *BMC Infect Dis*	GBD 2021; concentration and slope indices by SDI, projection to 2050	204 countries	DALY rate fell 32.4%; absolute inequality narrowed while relative inequality widened	No spatial analysis; no financing
Cao et al. (2026), *JAMA Dermatol*	GBD Foresight; skin NTD SDG index, 2020–2030	Global	Only 11 countries on track for a 75% DALY reduction for one skin NTD; higher government share of health spending linked to progress	Skin NTDs only; projected rather than observed
Theme C: Financing and health outcomes				

Study	Design and data	Scope	Key findings	Limitation relative to this study
Mishra & Newhouse (2009), *J Health Econ*	Cross-country panel, 1973–2004	118 countries	Doubling health aid per capita associated with 2% lower infant mortality	Infant mortality only; dated
Flaxman et al. (2010), *PLoS Med*	Bayesian model of net coverage and malaria aid, 2000–2008	44 African countries	Each US$1 per capita of malaria aid associated with +5.3 percentage points net ownership	Coverage, not burden
Akachi & Atun (2011), *PLoS One*	Panel regression and Lives Saved Tool, 2002–2008	34 SSA countries	High-burden countries received less aid per person at risk (US$3.90 vs US$7.05)	Child mortality only; pre-2010
Bendavid & Bhattacharya (2014), *JAMA Intern Med*	Panel with country and year fixed effects, 1974–2010	140 countries	US$1 billion in health aid associated with 364,800 fewer under-5 deaths; associations measurable for 3–5 years	All-cause outcomes; no NTDs
Hsiao & Emdin (2014), *J Epidemiol Glob Health*	Panel fixed effects and GMM	120 LMICs	Malaria aid associated with lower malaria mortality	No NTDs; no DALYs
Yan et al. (2015), *BMC Public Health*	Panel fixed effects, 1995–2010	Global Fund-eligible countries	+US$10 per capita associated with a 6.9% per year faster fall in under-5 malaria mortality	One donor; ends 2010

Study	Design and data	Scope	Key findings	Limitation relative to this study
Dolan et al. (2019), *PLoS One* [national]	Geospatial difference-in-differences, DHS	DRC	Bednet campaign reduced rural under-5 mortality by 41% in high-ecology areas	Single country and intervention
Weiss et al. (2022), *Popul Health Metr*	Synthetic control, USAID maternal and child health funding	High-USAID LMICs	Under-5 mortality 29 per 1,000 lower than synthetic control	Child mortality only
Sempungu et al. (2023), *Glob Public Health*	GBD 2019 joinpoint and two-way fixed effects, 2000–2017	Sub-Saharan Africa	Malaria DALY rate fell 47%; development assistance negatively associated with the malaria DALY rate	Malaria only; aid level rather than dependence share
Apeagyei et al. (2024b), *Malar J*	Malaria spending and stochastic frontier, 2000–2020	Endemic countries	Higher donor share of malaria spending associated with higher incidence efficiency	Efficiency, not burden
Gibson et al. (2025), *Lancet Glob Health*	Panel difference-in-differences of aid sanctions, 1990–2019	133 LMICs	Sanctions cut health aid by 17% and raised infant (3.1%), under-5 (3.6%) and maternal (6.4%) mortality	Maternal and child mortality only

Study	Design and data	Scope	Key findings	Limitation relative to this study
Shi et al. (2025), *China CDC Wkly*	Panel of donor funding and malaria cascade, 2011–2022	26 African countries	About 70% of malaria funding external; weak correlation with cascade coverage	Coverage, not burden
Theme D: Projections of the 2025 funding cuts				
Cavalcanti et al. (2025), *Lancet*	Fixed-effects Poisson panel (2001–2021) and microsimulation to 2030	133 countries	USAID funding associated with lower malaria (RR 0.49) and NTD (RR 0.50) mortality; cuts projected to cause 14.1 million extra deaths by 2030	One donor; mortality only; no country-level NTD projections
da Silva et al. (2026), *Lancet Glob Health*	Two-way fixed-effects Poisson (2002–2021) and microsimulation	93 LMICs	ODA associated with lower malaria (RR 0.44) and NTD (RR 0.46) mortality; severe defunding projected to cause 22.6 million extra deaths by 2030	Mortality only; aid level rather than dependence share; no between–within separation
Symons et al. (2025), *Lancet*	Geospatial transmission model, with and without PMI commodities	27 African countries	PMI averts 13.6 million cases and 104,000 deaths in 2025 (37.5% of malaria mortality in PMI areas)	Malaria only; one year; one donor

Study	Design and data	Scope	Key findings	Limitation relative to this study
Stover et al. (2025), *Lancet Glob Health*	Established mathematical models, 2025–2030	Up to 80 countries	Ending US funding: 4.1 million AIDS deaths, 606,900 TB deaths and 2.5 million other child deaths	No malaria or NTD outcomes
Hallett et al. (2025), *Lancet*	Epidemiological and costing models, 2027–2029	Global Fund-eligible countries	Optimal investment could save 23 million lives	HIV, TB and malaria only; investment case
Theme E: Domestic financing, aid displacement and donor transition				
Liese & Schubert (2009), *Int Health*	ODA commitments, 2003–2007	Donors	Only 0.6% of health aid went to NTDs (malaria 3.6%)	Descriptive; dated
Lordan et al. (2011), *Soc Sci Med*	Recipient-year aid panel	Aid recipients	HIV aid displaced malaria aid	Donor flows, not domestic spending
Shretta et al. (2017), *Malar J*	DAH and government expenditure tracking	35 eliminating countries	Malaria aid fell from US$176 million to US$62 million (2010–2013); domestic spending did not keep pace	Low-burden countries; descriptive
GBD Health Financing Collaborator Network (2019), *Lancet*	Spending by source, 1995–2016, projected to 2050	195 countries	Low-income countries held 0.4% of health spending but 10.0% of population	No burden outcomes

Study	Design and data	Scope	Key findings	Limitation relative to this study
Patenaude (2021), *Health Policy Plan*	System GMM, 2000–2015	134 countries	Aid had no significant effect on overall domestic public health spending	Under-5 mortality only
Huffstetler et al. (2022), *Health Policy Plan*	Scoping review of donor transitions (89 studies)	Middle-income countries	Unmanaged transitions produced shortfalls, stock-outs and service disruption	Qualitative
Apeagyei et al. (2025), *Lancet*	IHME DAH tracking and forecasts, 1990–2030	Global	DAH US$80.3 billion (2021), US$49.6 billion (2024), US$38.4 billion (2025, preliminary), forecast US$36.2 billion (2030)	Financing only; no health-loss estimates

Note. Findings are as reported in each study's abstract. [national] = landmark single-country study included for context. Thirty-two further eligible studies were not tabulated because they repeated the same GBD 2019–2021 descriptive design or were lower priority. DAH = development assistance for health; DHS = Demographic and Health Surveys; GMM = generalised method of moments; LMICs = low- and middle-income countries; MAP = Malaria Atlas Project; ODA = official development assistance; PMI = US President's Malaria Initiative; RR = risk ratio; SDI = Socio-demographic Index; SSA = sub-Saharan Africa.

What the existing evidence shows

Global burden is falling, but published analyses stop at GBD 2021.

- Across 204 countries, combined NTD and malaria DALYs fell by about 18%, from 87.4 million in 1990 to 71.6 million in 2021 (Zheng et al., 2025b).
- The age-standardised DALY rate fell from 1,506.5 to 1,020.3 per 100,000 over the same period (Lin et al., 2025).

- WHO reports 1.495 billion people requiring NTD interventions in 2023, 32% fewer than in 2010, and 282 million malaria cases and 610,000 deaths in 2024.
- The GBD 2023 capstone reports a 25.8% fall in age-standardised DALYs from communicable, maternal, neonatal and nutritional causes since 2010, and warns that aid cuts threaten these gains (GBD 2023 Diseases and Injuries Collaborators, 2025).
- No retrieved study reports GBD 2023 estimates for the NTD and malaria group as a whole. GBD 2023 work covers single diseases (Chagas disease) or deaths in five BRICS countries.

Burden is concentrated in low-SDI countries, and the relative gap is not closing.

- Low- and lower-middle-income countries held 69% of NTD burden in 1990 and 81% in 2010 (Stolk et al., 2016).
- With GBD 2019 data, the concentration index by Human Development Index was −0.69, the most pro-poor of all communicable-disease groups (Emadi et al., 2021).
- GBD 2021 analyses agree that absolute inequality narrowed while relative inequality widened (Feng et al., 2025; Lin et al., 2025). Western sub-Saharan Africa alone held 51.2% of the global burden (Lin et al., 2025).
- These studies rank countries by SDI or income. None decomposes inequality into between-region and within-region components, and none tests whether burden clusters geographically.

Spatial evidence is limited to single diseases or to malaria in Africa.

- The only cross-country spatial-econometric study found significant Moran's I clustering of malaria cases and deaths in 38 sub-Saharan African countries in every year from 2010 to 2022. It also found aid per capita to be positively associated with burden, which is consistent with donors targeting high-burden countries (Namubiru, 2025).
- Geostatistical NTD studies model the prevalence of single diseases, such as soil-transmitted helminthiasis (Khaki et al., 2025).
- No study maps global clusters of combined NTD and malaria DALYs.

Aid is consistently associated with lower mortality, but only with mortality.

- Cross-country panels associate health aid with lower infant and under-5 mortality (Mishra & Newhouse, 2009; Bendavid & Bhattacharya, 2014). Malaria-specific aid has been associated with higher net coverage and lower malaria mortality (Flaxman et al., 2010; Hsiao & Emdin, 2014; Yan et al., 2015).
- The two largest recent panels used two-way fixed-effects Poisson models. USAID funding across 133 countries was associated with 51% lower malaria mortality and 50% lower NTD mortality (Cavalcanti et al., 2025). Total ODA across 93 low- and middle-

income countries was associated with 56% lower malaria mortality and 54% lower NTD mortality (da Silva et al., 2026).

- All of these studies:
  - model aid volume rather than the share of health spending financed externally;
  - report mortality, although most NTD burden is non-fatal disability;
  - do not separate between-country differences from within-country change.
- Several studies note reverse allocation, with aid flowing to where burden is highest (Akachi & Atun, 2011; Namubiru, 2025). This makes between-country associations hard to interpret.

Projections of the 2025 cuts cover deaths, HIV, TB and malaria, but not NTD DALYs.

- Global development assistance for health fell from US$80.3 billion in 2021 to US$49.6 billion in 2024, and to a preliminary US$38.4 billion in 2025 (Apeagyei et al., 2025). WHO reports a 41% fall in NTD-targeted ODA between 2018 and 2023.
- Projections of the consequences cover:
  - all-cause deaths: 14.1 million (Cavalcanti et al., 2025) and 22.6 million (da Silva et al., 2026) extra deaths by 2030;
  - HIV, TB and child deaths (Stover et al., 2025);
  - a single year of PMI-supported malaria control (Symons et al., 2025).
- A targeted search found no study projecting NTD DALYs, country by country, under the cuts.

There is little evidence that governments replace lost aid.

- Aid has not been shown to raise overall domestic public health spending (Patenaude, 2021).
- Malaria-eliminating countries did not increase domestic spending when aid fell (Shretta et al., 2017).
- NTDs received only 0.6% of health aid even at its peak (Liese & Schubert, 2009).
- Unmanaged donor transitions have produced stock-outs and service disruption (Huffstetler et al., 2022).

Gaps this study addresses

The search identified seven gaps. Our study addresses six of them directly and the seventh in part.

Table 2

Gaps in the Existing Evidence and How This Study Addresses Them

Gap	Evidence from the search	How this study addresses it
1. No GBD 2023 analysis of combined NTD and malaria burden	All 204-country trend and inequality studies use GBD 2019 or 2021; GBD 2023 work covers Chagas disease, five BRICS countries or all causes	First analysis of the NTD and malaria group using GBD 2023, 1990–2023, at global, super-regional and national level
2. Spatial clustering of NTD and malaria DALYs untested	Only one spatial study, of malaria cases and deaths in 38 sub-Saharan African countries; NTD geostatistics cover single-disease prevalence	Global Moran's I and FDR-corrected LISA for NTD and malaria DALYs in 203 countries, robust to four weight specifications
3. Inequality not decomposed	Concentration and slope indices by SDI are reported, but no Theil decomposition in any of 76 eligible studies	Concentration index by SDI plus Theil decomposition into between- and within-super-region inequality
4. Financing dependence not studied alongside burden	Aid studies model aid volume; none uses the external share of health expenditure; none maps where burden and dependence coincide	Bivariate classification of burden and external share, identifying 34 high-burden, high-dependence countries
5. Between-country and within-country associations conflated	Panels use pooled or fixed-effects designs without within–between comparison; reverse allocation documented	Seven estimators, including cross-section, two-way fixed effects, Mundlak, first differences and spatial panels, with formal tests and missing-data sensitivity analyses
6. NTD outcomes measured as mortality only	USAID and ODA studies report NTD mortality risk ratios, although NTD burden is mostly disability	DALYs as the outcome, capturing non-fatal NTD burden

Gap	Evidence from the search	How this study addresses it
7. No country-level NTD DALY projections under the 2025 cuts	Projections cover all-cause, HIV, TB, child and malaria deaths; targeted search found no NTD projection	Conditional scenario estimates of excess NTD and malaria DALYs by country and super-region, with a domestic financing-gap analysis (addressed in part: estimates rest on an attribution assumption)

Note. FDR = false-discovery rate; LISA = local indicators of spatial association; ODA = official development assistance; SDI = Socio-demographic Index.

Research in context (rewritten)

This is draft text for the manuscript's Research in Context panel, written in Lancet style.

Evidence before this study

We searched PubMed from database inception to Oct 21, 2026, with no language restrictions. The search terms were "neglected tropical disease*", "malaria", "Global Burden of Disease", "disability-adjusted life years", "inequalit*", "concentration index", "spatial", "Moran", "development assistance for health", "donor funding", "USAID" and "funding cuts", in six combined strings. We also reviewed WHO and IHME reports.

Of 704 unique records, 76 studies met our inclusion criteria. Analyses of GBD 2019 and 2021 data showed that combined NTD and malaria DALYs fell by about 18% between 1990 and 2021. Over the same period, burden became increasingly concentrated in low-SDI countries, especially in western sub-Saharan Africa, and relative inequality widened as absolute inequality narrowed. No study had used GBD 2023 estimates for the NTD and malaria group, tested the geographical clustering of these DALYs across countries, or decomposed inequality between and within regions.

Cross-country panel studies consistently associated development assistance with lower malaria, NTD and child mortality. The largest, of USAID funding in 133 countries and of total aid in 93 countries, projected 14 to 23 million additional deaths by 2030 from current funding cuts. These studies modelled aid volume rather than national dependence on external financing, reported mortality although NTD burden is mostly non-fatal, and did not distinguish between-country differences from within-country change. Other projections of the 2025 cuts addressed HIV, tuberculosis, child mortality and one year of malaria control. We found no country-level estimates of NTD DALYs under the cuts.

Added value of this study

To our knowledge, this is the first analysis of NTD and malaria burden using GBD 2023 that brings together spatial clustering, socioeconomic inequality and national dependence on external health financing for 204 countries and territories.

We found:

- strong geographical clustering of burden, with sub-Saharan Africa holding 80.2% of combined DALYs;
- concentration of burden in low-SDI countries;
- inequality that arose mainly between super-regions;
- 34 countries where high NTD burden coincides with high external financing dependence, together holding 27.2% of global NTD DALYs and 52.3% of malaria DALYs.

By comparing seven panel estimators, we showed that the inverse association between external financing share and NTD burden seen between countries was not detectable within countries over time. The malaria association depended on model specification. We also provide conditional, country-level estimates of DALYs at risk under specified funding-cut scenarios.

Implications of all the available evidence

NTD and malaria burden remains heavily concentrated where financing dependence is greatest, so the 2025 reductions in development assistance fall disproportionately on the most affected populations. Protecting continuity of services in high-burden, high-dependence countries is supported by the convergence of burden, inequality and financing evidence. However, aggregate external-financing shares should not be read as disease-specific effects. Estimating the true consequences of the cuts requires programme-level disbursement data linked to coverage and disease-specific outcomes, and domestic resource mobilisation planned alongside disease priorities.

How the evidence should reshape our global, regional and national analysis

The literature points to twelve revisions, listed in Table 3. Four are essential for positioning the paper against existing work:

- reporting absolute and relative inequality over time;
- disaggregating sub-Saharan Africa into its four GBD regions;
- separating disability from mortality in NTD burden;
- correcting the two scenario calculations that the manuscript already flags as provisional.

Table 3

Recommended Revisions to the Global, Regional and National Analyses, With the Evidence Motivating Each

Level	Revision	Evidence motivating the revision	Priority
Global	Benchmark 2023 results against the last published GBD 2021 figures: 71.6 million DALYs and an age-standardised rate of 1,020.3 per 100,000 in 2021, against our 70.8 million in 2023. State that GBD rounds re-estimate the full series, so the figures are not a direct change	Zheng et al. (2025b); Lin et al. (2025)	Essential
Global	Report absolute (slope index of inequality) and relative (concentration index) inequality for 1990, 2010 and 2023, not 2023 alone	Feng et al. (2025) and Lin et al. (2025) show absolute inequality narrowing while relative inequality widens; our single-year index cannot test this	Essential
Global	Split NTD DALYs into years lived with disability (YLDs) and years of life lost (YLLs), and report the YLD share	The aid–outcome literature measures NTD mortality only (Cavalcanti et al., 2025; da Silva et al., 2026); showing that most NTD burden is disability justifies our DALY outcome	Essential
Global	Compare the 2023 GBD burden with WHO indicators: people requiring NTD interventions (1.495 billion in 2023) and malaria cases and deaths (282 million and 610,000 in 2024)	WHO global reports; triangulation strengthens the GBD-based findings	Recommended

Level	Revision	Evidence motivating the revision	Priority
Regional	Disaggregate sub-Saharan Africa into its four GBD regions (western, central, eastern, southern) and report the 21 GBD regions in the supplement	Lin et al. (2025) found western sub-Saharan Africa alone holds 51.2% of burden; the super-region hides this	Essential
Regional	Extend the Theil decomposition to regions within super-regions, and repeat it for 2010 and 2023	No study decomposes inequality; showing change over time strengthens our original contribution	Recommended
Regional	Repeat LISA for 2010 and 2023 to show whether hotspots persisted, expanded or contracted	Namubiru (2025) found persistent malaria clustering in Africa; temporal cluster maps would extend this to NTDs and globally	Recommended
National	Add country rankings, top 10 by count and by age-standardised rate, with 2010–2023 change and uncertainty intervals, and flag countries where burden rose	GBD 2019 analyses report rising NTD incidence in some settings (Lin et al., 2022); national rankings make results actionable	Essential
National	Report cause-specific national leaders (for example, lymphatic filariasis in South Asia, schistosomiasis and onchocerciasis in sub-Saharan Africa, Chagas disease in Latin America)	Aggregate NTD estimates conceal disease-specific geography and response times (Borlase et al., 2022)	Recommended
National – financing	Add aid volume (development assistance per capita) as an alternative exposure, and all-cause or malaria mortality as a sensitivity outcome	Allows direct comparison with the mortality risk ratios of Cavalcanti et al. (2025) and da Silva et al. (2026)	Recommended

Level	Revision	Evidence motivating the revision	Priority
National – financing	Present the between-country and within-country (Mundlak) estimates side by side as a headline finding	Reverse allocation of aid to high-burden countries (Akachi & Atun, 2011; Namubiru, 2025) explains why the between-country estimate is not interpretable alone	Essential
National – scenarios	Correct the two issues the manuscript flags: (1) supply 2010 baselines for the five NTD causes that have only 2023 data (Ebola, Guinea worm disease, yellow fever, Zika virus, other NTDs), or restrict the accounting model to the 14 causes with both years; (2) convert the malaria log(1 + rate) coefficient correctly to rate and count changes	Current excess-DALY estimates are provisional; projections are only comparable with published estimates once corrected	Essential

Note. Essential revisions are needed to position the paper against existing studies or to correct calculations. Recommended revisions strengthen the contribution but are not required for internal validity.

References

All article fields come from PubMed records retrieved on Oct 21, 2026. The references are listed alphabetically because this working document uses author–year citations; they will be renumbered in Vancouver order when moved into the manuscript.

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from a secondary summary of the WHO report, and the WHO 2026 World NTD Day page gives 58. Check both against the report PDF before citing either.